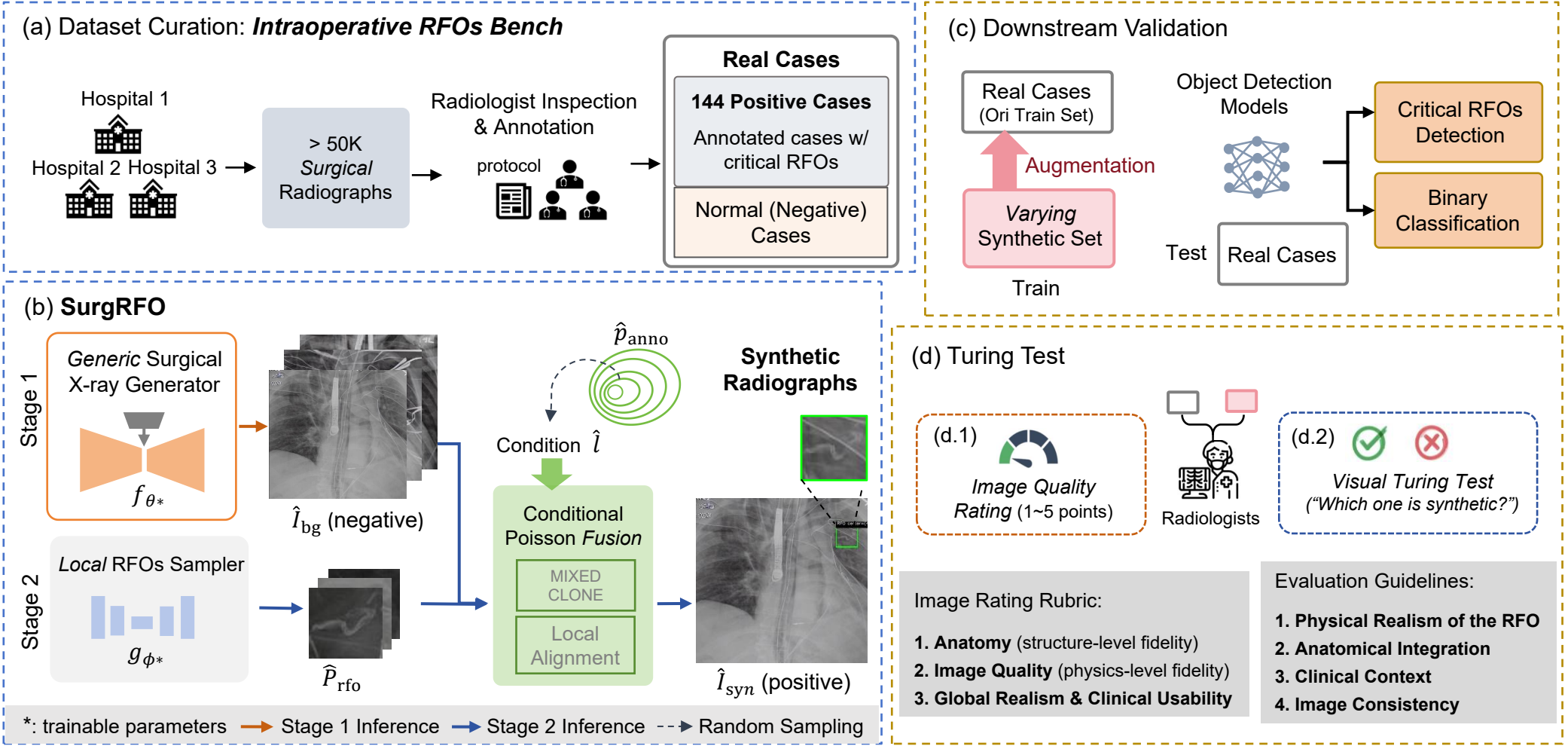
(a) Dataset Curation: Intraoperative RFOs Bench
Hospital 1
Hospital 2
Hospital 3
> 50K Surgical Radiographs
Radiologist Inspection & Annotation
protocol
Real Cases
144 Positive Cases
Annotated cases w/ critical RFOs
Normal (Negative) Cases
(b) SurgRFO
Stage 1
Generic Surgical X-ray Generator
$f_{\theta*}$
$\hat{I}_{bg}$ (negative)
Stage 2
Local RFOs Sampler
$g_{\phi*}$
$\hat{P}_{rfo}$
$\hat{p}_{anno}$
Condition $\hat{l}$
Conditional Poisson Fusion
MIXED CLONE
Local Alignment
Synthetic Radiographs
$\hat{I}_{syn}$ (positive)
*: trainable parameters
Stage 1 Inference
Stage 2 Inference
Random Sampling
(c) Downstream Validation
Real Cases (Ori Train Set)
Augmentation
Varying Synthetic Set
Train
Object Detection Models
Critical RFOs Detection
Binary Classification
Test
Real Cases
(d) Turing Test
(d.1)
Image Quality Rating (1~5 points)
Radiologists
(d.2)
Visual Turing Test
("Which one is synthetic?")
Image Rating Rubric:
1. Anatomy (structure-level fidelity)
2. Image Quality (physics-level fidelity)
3. Global Realism & Clinical Usability
Evaluation Guidelines:
1. Physical Realism of the RFO
2. Anatomical Integration
3. Clinical Context
4. Image Consistency